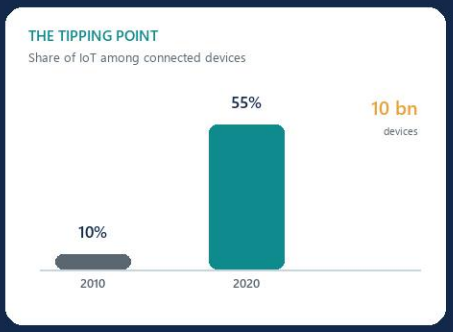
ACADEMIC INFOGRAPHIC · 26CAP-608

Internet of Things (IoT): Societal, Scientific, and Technical Foundations

Societal challenges, technical constraints and research fields for a sustainable IoT



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01 THE PROBLEM

~10 bn

IoT devices installed worldwide by 2020, a tenfold decade of growth.

50%+

of all connected devices are now IoT, so failures reach the physical world.

0.001%

of a conventional memory budget is available on some constrained nodes.

02 CAUSES → CONSEQUENCES

ROOT CAUSES	CONSEQUENCES
1 **Resource limits** — MicroWatt energy, kilobytes of RAM, slow CPUs	1 **Weak defences** — Conventional cryptography does not fit
2 **Insecure updates** — Long-lived firmware without signed manifests	2 **Cyber-physical harm** — Compromised devices can actuate the real world
3 **Pervasive sensing** — Continuous collection of behavioural data	3 **Privacy erosion** — Surveillance and behavioural monetisation
4 **Fragmented standards** — Vendor lock-in and non-interoperable stacks	4 **E-waste and dependency** — Short lifetimes and foreign supply chains

03 SUSTAINABLE & TECHNICAL SOLUTIONS

1 Edge Computing

Process telemetry near the source to cut latency, bandwidth and exposure of raw data.

2 Lightweight Cryptography

Use ChaCha20, AES-128 and elliptic-curve primitives sized for constrained MCUs.

3 Federated Learning

Train models locally and share only aggregated updates to preserve privacy.

4 Open Standards & RIOT OS

Build on vendor-independent open source and standard protocols to avoid lock-in.

Open standards · vendor independence

04 ACTION PLAN BY STAKEHOLDER

Governments & Regulators	Enforce privacy law, fund open standards and require security-by-design procurement.
Academic & Research Institutions	Develop lightweight security, benchmark energy use and publish reproducible results.
Developers & Engineers	Adopt secure update pipelines, least privilege and energy-aware programming.
End Users & Consumers	Demand transparency, maintain devices and prefer products with long support lifetimes.

KEY TAKEAWAY

Digital sovereignty and secure-by-design engineering are not optional features: they are the conditions under which a sustainable IoT can earn and retain public trust.